

Machine Learning-Based Crop Recommendation System

Alusine Bah

Electrical & Electronics Engineering
Islamic University of Technology (IUT), OIC

September 27, 2026

Abstract

Agriculture remains one of the most important sectors for sustaining human life, and the choice of the appropriate crop for a given plot of land is critical to maximizing productivity and minimizing waste. This project presents a machine learning-based Crop Recommendation System that predicts the most suitable crop to grow based on soil nutrient levels and climatic conditions. The system is trained on a dataset containing nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, soil pH, and rainfall. Five classification algorithms were trained and compared: Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, and Support Vector Machine. The Random Forest classifier achieved the best performance, with a test accuracy of 99.32%. The trained model was deployed as an interactive web application using Gradio, allowing users to enter soil and climate parameters and receive an instant crop recommendation.

1 Introduction

Agriculture is a central part of the economy of most nations. When deciding which crop to plant, farmers have traditionally relied on past experience and local knowledge. However, changing climate patterns and rising food demand have made it increasingly important to adopt modern, data-driven methods for agricultural decision-making.

Machine learning offers a practical solution for analyzing agricultural data and generating accurate predictions. This project develops a Crop Recommendation System that uses supervised learning to recommend the most suitable crop based on soil and environmental inputs.

2 Problem Statement

Farmers face several challenges when selecting which crop to grow:

- Limited knowledge of soil nutrient composition.
- Uncertainty in weather conditions.
- Inefficient use of water and fertilizer.
- Reduced yield due to poor crop selection.

This project aims to build a system that analyzes soil and environmental data and returns accurate crop recommendations, reducing the guesswork involved in the decision.

3 Objectives

The main objectives of this project are:

1. To build a machine learning model that predicts the most suitable crop from soil and climate inputs.
2. To analyze the relationship between soil nutrients, climate parameters, and crop suitability.
3. To improve agricultural decision-making through an accessible, easy-to-use interface.

4 Dataset Description

The dataset used in this project contains the agricultural parameters that determine crop growth. Each row represents a specific combination of soil and environmental conditions, along with the recommended crop label.

4.1 Features

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)
- Temperature ($^{\circ}\text{C}$)
- Humidity (%)
- Soil pH
- Rainfall (mm)

4.2 Target

The target variable is the crop label — the most appropriate crop to plant given the associated conditions. The dataset contains 22 distinct crop classes.

4.3 Dataset Summary

Property	Value
Total records	2,200
Number of features	7
Number of classes	22
Samples per class	100
Missing values	0

Table 1: Summary of the dataset used in this project.

The 22 crop classes are: rice, maize, chickpea, kidneybeans, pigeonpeas, mothbeans, mungbean, blackgram, lentil, pomegranate, banana, mango, grapes, watermelon, muskmelon, apple, orange, papaya, coconut, cotton, jute, and coffee.

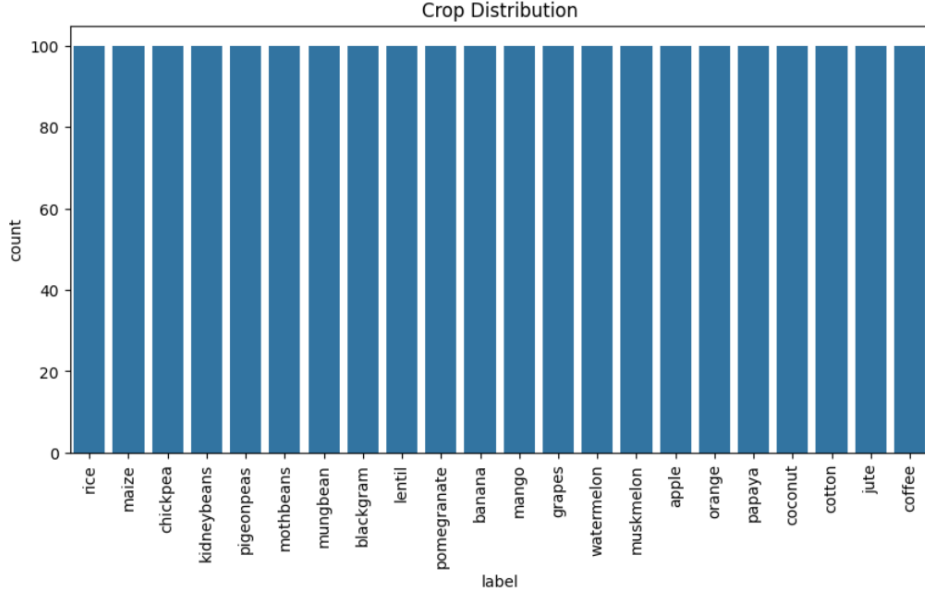


Figure 1: Distribution of samples across the 22 crop classes. Each class contains exactly 100 samples.

5 Methodology

The project was carried out in a structured sequence of steps, from data loading to model deployment.

5.1 Data Loading

The dataset was loaded into Python using the Pandas library. Basic checks confirmed the shape, data types, and absence of missing values.

5.2 Data Preprocessing

- Verified there were no missing values.
- Confirmed appropriate data types for all columns.
- Separated the input features (X) from the target variable (y).
- Applied `StandardScaler` to normalize feature ranges. The scaler was fit on the training set only, and then applied to the test set to avoid data leakage.

5.3 Train-Test Split

The dataset was divided into 80% training data and 20% test data using `train_test_split` with `random_state=42`. This resulted in 1,760 training samples and 440 test samples.

5.4 Model Training

Five classification algorithms were trained and evaluated on the same split:

- Logistic Regression (`max_iter=5000`)
- Decision Tree Classifier

- Random Forest Classifier
- K-Nearest Neighbors
- Support Vector Machine (RBF kernel)

5.5 Model Evaluation

Each model was evaluated using classification accuracy on the held-out test set. The best-performing model was selected for deployment.

6 Model Selection

Table 2 summarizes the test accuracy obtained by each model.

Model	Test Accuracy
Logistic Regression	96.36%
K-Nearest Neighbors	95.68%
Support Vector Machine	96.82%
Decision Tree	98.86%
Random Forest	99.32%

Table 2: Comparison of model accuracies on the test set.

The Random Forest classifier achieved the highest accuracy. Its advantage over a single Decision Tree comes from bagging — training many trees on different subsets of the data and averaging their predictions — which reduces the variance to which a single tree is prone.

Based on these results, Random Forest was selected as the final model for the system.

7 System Implementation

The system was implemented in Python and developed in Google Colab. The implementation steps were:

1. Load and explore the dataset.
2. Preprocess the data and split into training and test sets.
3. Train the five candidate models.
4. Select the best-performing model.
5. Build a prediction function around the trained model.
6. Wrap the function in an interactive interface for end users.

8 System Interface

A simple interactive interface was built using the Gradio library. The user enters seven numeric values:

- Nitrogen
- Phosphorus

- Potassium
- Temperature
- Humidity
- Soil pH
- Rainfall

The trained model processes these values (after applying the same **StandardScaler**) and returns the recommended crop. The interface is designed to be usable without any technical background.

Crop Recommendation System

Enter soil and weather parameters to predict the best crop.

Nitrogen: 0

Phosphorus: 0

Potassium: 0

Temperature: 0

Humidity: 0

pH: 0

Rainfall: 0

output:

Flag

Clear Submit

Use via API Built with Gradio Settings

Figure 2: The Gradio interface input form. The user enters seven soil and climate parameters.

Crop Recommendation System

Enter soil and weather parameters to predict the best crop.

Nitrogen: 90

Phosphorus: 42

Potassium: 43

Temperature: 22

Humidity: 82

pH: 6.5

Rainfall: 220

output: rice

Flag

Clear Submit

Figure 3: The model returns the recommended crop **rice** for realistic rice-growing conditions.

9 Results

The system successfully predicts crops from the input parameters, with the Random Forest model producing reliable recommendations across all 22 classes. For example, given the inputs

N=90, P=42, K=43, temp=22, humidity=82, ph=6.5, rainfall=220

the model returns **rice**, which is agronomically consistent with the warm, humid, high-rainfall conditions represented by those values.

10 Advantages and Limitations

Advantages

- Easy to use, with a clear input form and instant output.
- Fast predictions suitable for practical use.
- Removes reliance on guesswork in crop selection.
- Can contribute to improved agricultural productivity.

Limitations

- The quality of recommendations depends on the quality of the dataset.
- Weather data is not retrieved in real time — values must be entered manually.
- Predictions are only reliable for inputs within the ranges seen during training.
- The model has not been validated on independent farm trials.

11 Future Improvements

Future work on this system could include:

1. Integration of live weather data.
2. Development of a mobile application.
3. Expansion of the dataset with regional and seasonal variation.
4. Exploration of more advanced machine learning and deep learning models.
5. Support for multiple languages.

12 Conclusion

The Crop Recommendation System demonstrates how machine learning can be effectively applied to agriculture. By analyzing soil nutrients and environmental conditions, the system provides accurate crop recommendations that can help farmers make better decisions, increase yields, and reduce waste. The project highlights the potential of data-driven methods to address real agricultural problems and to support sustainable farming practices.